

1. Lexical analysis, tokens and regular expressions — Concept foundation

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Examiner perspective. A strong question on Lexical analysis, tokens and regular expressions can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Lexical analysis, tokens and regular expressions. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

FIRST/FOLLOW drive predictive parsing

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Topic summary. By the end of this section you should be able to define the core terms of Lexical analysis, tokens and regular expressions, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.

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Derivation mindset. Start from the definition, state the invariant, perform one legal transformation at a time, and finish by checking the result against a limiting case. For Lexical analysis, tokens and regular expressions, this is more useful than memorizing a proof verbatim because GATE can alter the input representation while preserving the underlying theorem or invariant.

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Exam workflow. Read the stem once for the model and once for the requested quantity. Extract symbols and constraints, classify the question, eliminate impossible choices, calculate only what is necessary, then perform a sanity check. For Lexical analysis, tokens and regular expressions, a 20–40 second classification step often saves several minutes of algebra.

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Edge-case checklist. Test zero/one-sized input, empty structures, repeated values, maximum/minimum representable values, ties, disconnected components, overflow/underflow, and invalid assumptions whenever they make sense for Lexical analysis, tokens and regular expressions. Edge cases are where apparently correct reasoning often breaks.

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Formula sheet. Relevant formulas and relationships include: FIRST/FOLLOW drive predictive parsing; LR shift/reduce decisions use item states; Three-address code uses temporaries; Live variable: variable is live if used on some future path; Available expression must be computed on every predecessor path. Do not treat these as isolated equations. Write the meaning and units of every variable, and note when the equation stops being valid. A formula earns marks only when the model behind it is correct.

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For **Finite automata for lexical recognition**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

The central idea is to build a chain from representation to operation to consequence. In **Finite automata for lexical recognition**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Derivation mindset. Start from the definition, state the invariant, perform one legal transformation at a time, and finish by checking the result against a limiting case. For Finite automata for lexical recognition, this is more useful than memorizing a proof verbatim because GATE can alter the input representation while preserving the underlying theorem or invariant.

Worked reasoning. Consider a small exam-sized instance related to Finite automata for lexical recognition. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

FIRST/FOLLOW drive predictive parsing

GATE traps. Common mistakes around Finite automata for lexical recognition include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

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2. Finite automata for lexical recognition — Problem-solving workflow

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Exam workflow. Read the stem once for the model and once for the requested quantity. Extract symbols and constraints, classify the question, eliminate impossible choices, calculate only what is necessary, then perform a sanity check. For Finite automata for lexical recognition, a 20–40 second classification step often saves several minutes of algebra.

Worked reasoning. Consider a small exam-sized instance related to Finite automata for lexical recognition. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

LR shift/reduce decisions use item states

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2. Finite automata for lexical recognition — Edge cases

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Edge-case checklist. Test zero/one-sized input, empty structures, repeated values, maximum/minimum representable values, ties, disconnected components, overflow/underflow, and invalid assumptions whenever they make sense for Finite automata for lexical recognition. Edge cases are where apparently correct reasoning often breaks.

Worked reasoning. Consider a small exam-sized instance related to Finite automata for lexical recognition. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Three-address code uses temporaries

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2. Finite automata for lexical recognition — Formula & memory map

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Formula sheet. Relevant formulas and relationships include: FIRST/FOLLOW drive predictive parsing; LR shift/reduce decisions use item states; Three-address code uses temporaries; Live variable: variable is live if used on some future path; Available expression must be computed on every predecessor path. Do not treat these as isolated equations. Write the meaning and units of every variable, and note when the equation stops being valid. A formula earns marks only when the model behind it is correct.

Worked reasoning. Consider a small exam-sized instance related to Finite automata for lexical recognition. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

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2. Finite automata for lexical recognition — Unit synthesis

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Examiner perspective. A strong question on Finite automata for lexical recognition can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Finite automata for lexical recognition. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Available expression must be computed on every predecessor path

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3. Parsing foundations and ambiguity — Concept foundation

For **Parsing foundations and ambiguity**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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High-yield formula / relation
FIRST/FOLLOW drive predictive parsing

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3. Parsing foundations and ambiguity — Deep dive

For **Parsing foundations and ambiguity**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Examiner perspective. A strong question on Parsing foundations and ambiguity can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Parsing foundations and ambiguity. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

LR shift/reduce decisions use item states

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3. Parsing foundations and ambiguity — Worked example

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Worked reasoning. Consider a small exam-sized instance related to Parsing foundations and ambiguity. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation
Three-address code uses temporaries

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High-yield formula / relation

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3. Parsing foundations and ambiguity — Comparison & intuition

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Comparison table. Ask what happens when the input is small versus large, structured versus arbitrary, and best-case versus worst-case. Then compare the primary method for Parsing foundations and ambiguity with its nearest alternative. The winning method is not always the asymptotically fastest one; constraints, preprocessing, memory and correctness conditions matter.

Worked reasoning. Consider a small exam-sized instance related to Parsing foundations and ambiguity. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

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Formula sheet. Relevant formulas and relationships include: FIRST/FOLLOW drive predictive parsing; LR shift/reduce decisions use item states; Three-address code uses temporaries; Live variable: variable is live if used on some future path; Available expression must be computed on every predecessor path. Do not treat these as isolated equations. Write the meaning and units of every variable, and note when the equation stops being valid. A formula earns marks only when the model behind it is correct.

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Examiner perspective. A strong question on Parsing foundations and ambiguity can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Parsing foundations and ambiguity. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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Available expression must be computed on every predecessor path

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4. LL, LR, SLR and LALR parsing — Concept foundation

For **LL, LR, SLR and LALR parsing**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Examiner perspective. A strong question on LL, LR, SLR and LALR parsing can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to LL, LR, SLR and LALR parsing. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

FIRST/FOLLOW drive predictive parsing

GATE traps. Common mistakes around LL, LR, SLR and LALR parsing include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

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High-yield formula / relation

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High-yield formula / relation
Three-address code uses temporaries

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Comparison table. Ask what happens when the input is small versus large, structured versus arbitrary, and best-case versus worst-case. Then compare the primary method for LL, LR, SLR and LALR parsing with its nearest alternative. The winning method is not always the asymptotically fastest one; constraints, preprocessing, memory and correctness conditions matter.

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4. LL, LR, SLR and LALR parsing — Derivation / proof idea

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Derivation mindset. Start from the definition, state the invariant, perform one legal transformation at a time, and finish by checking the result against a limiting case. For LL, LR, SLR and LALR parsing, this is more useful than memorizing a proof verbatim because GATE can alter the input representation while preserving the underlying theorem or invariant.

Worked reasoning. Consider a small exam-sized instance related to LL, LR, SLR and LALR parsing. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

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4. LL, LR, SLR and LALR parsing — Problem-solving workflow

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Exam workflow. Read the stem once for the model and once for the requested quantity. Extract symbols and constraints, classify the question, eliminate impossible choices, calculate only what is necessary, then perform a sanity check. For LL, LR, SLR and LALR parsing, a 20–40 second classification step often saves several minutes of algebra.

Worked reasoning. Consider a small exam-sized instance related to LL, LR, SLR and LALR parsing. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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4. LL, LR, SLR and LALR parsing — Edge cases

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Edge-case checklist. Test zero/one-sized input, empty structures, repeated values, maximum/minimum representable values, ties, disconnected components, overflow/underflow, and invalid assumptions whenever they make sense for LL, LR, SLR and LALR parsing. Edge cases are where apparently correct reasoning often breaks.

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High-yield formula / relation
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4. LL, LR, SLR and LALR parsing — Formula & memory map

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Formula sheet. Relevant formulas and relationships include: FIRST/FOLLOW drive predictive parsing; LR shift/reduce decisions use item states; Three-address code uses temporaries; Live variable: variable is live if used on some future path; Available expression must be computed on every predecessor path. Do not treat these as isolated equations. Write the meaning and units of every variable, and note when the equation stops being valid. A formula earns marks only when the model behind it is correct.

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4. LL, LR, SLR and LALR parsing — Unit synthesis

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5. Syntax-directed translation and semantic analysis — Concept foundation

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High-yield formula / relation

Available expression must be computed on every predecessor path

GATE traps. Common mistakes around Syntax-directed translation and semantic analysis include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

Topic summary. By the end of this section you should be able to define the core terms of Syntax-directed translation and semantic analysis, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.

5. Syntax-directed translation and semantic analysis — Derivation / proof idea

For **Syntax-directed translation and semantic analysis**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

The central idea is to build a chain from representation to operation to consequence. In **Syntax-directed translation and semantic analysis**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Derivation mindset. Start from the definition, state the invariant, perform one legal transformation at a time, and finish by checking the result against a limiting case. For Syntax-directed translation and semantic analysis, this is more useful than memorizing a proof verbatim because GATE can alter the input representation while preserving the underlying theorem or invariant.

Worked reasoning. Consider a small exam-sized instance related to Syntax-directed translation and semantic analysis. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

FIRST/FOLLOW drive predictive parsing

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5. Syntax-directed translation and semantic analysis — Problem-solving workflow

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Exam workflow. Read the stem once for the model and once for the requested quantity. Extract symbols and constraints, classify the question, eliminate impossible choices, calculate only what is necessary, then perform a sanity check. For Syntax-directed translation and semantic analysis, a 20–40 second classification step often saves several minutes of algebra.

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High-yield formula / relation

LR shift/reduce decisions use item states

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Edge-case checklist. Test zero/one-sized input, empty structures, repeated values, maximum/minimum representable values, ties, disconnected components, overflow/underflow, and invalid assumptions whenever they make sense for Syntax-directed translation and semantic analysis. Edge cases are where apparently correct reasoning often breaks.

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High-yield formula / relation

Three-address code uses temporaries

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5. Syntax-directed translation and semantic analysis — Formula & memory map

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Formula sheet. Relevant formulas and relationships include: FIRST/FOLLOW drive predictive parsing; LR shift/reduce decisions use item states; Three-address code uses temporaries; Live variable: variable is live if used on some future path; Available expression must be computed on every predecessor path. Do not treat these as isolated equations. Write the meaning and units of every variable, and note when the equation stops being valid. A formula earns marks only when the model behind it is correct.

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Examiner perspective. A strong question on Syntax-directed translation and semantic analysis can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Syntax-directed translation and semantic analysis. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

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6. Intermediate representations and three-address code — Concept foundation

For **Intermediate representations and three-address code**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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High-yield formula / relation

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6. Intermediate representations and three-address code — Deep dive

For **Intermediate representations and three-address code**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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High-yield formula / relation

LR shift/reduce decisions use item states

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6. Intermediate representations and three-address code — Worked example

For **Intermediate representations and three-address code**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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High-yield formula / relation
Three-address code uses temporaries

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6. Intermediate representations and three-address code — GATE traps

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High-yield formula / relation

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6. Intermediate representations and three-address code — Comparison & intuition

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Comparison table. Ask what happens when the input is small versus large, structured versus arbitrary, and best-case versus worst-case. Then compare the primary method for Intermediate representations and three-address code with its nearest alternative. The winning method is not always the asymptotically fastest one; constraints, preprocessing, memory and correctness conditions matter.

Worked reasoning. Consider a small exam-sized instance related to Intermediate representations and three-address code. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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6. Intermediate representations and three-address code — Derivation / proof idea

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Examiner perspective. A strong question on Intermediate representations and three-address code can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Intermediate representations and three-address code. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Available expression must be computed on every predecessor path

GATE traps. Common mistakes around Intermediate representations and three-address code include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

Topic summary. By the end of this section you should be able to define the core terms of Intermediate representations and three-address code, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.

7. Runtime environments, activation records and symbol tables — Concept foundation

For **Runtime environments, activation records and symbol tables**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

The central idea is to build a chain from representation to operation to consequence. In **Runtime environments, activation records and symbol tables**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Examiner perspective. A strong question on Runtime environments, activation records and symbol tables can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Runtime environments, activation records and symbol tables. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

FIRST/FOLLOW drive predictive parsing

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7. Runtime environments, activation records and symbol tables — Deep dive

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High-yield formula / relation

LR shift/reduce decisions use item states

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7. Runtime environments, activation records and symbol tables — Worked example

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High-yield formula / relation

Three-address code uses temporaries

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High-yield formula / relation

Live variable: variable is live if used on some future path

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7. Runtime environments, activation records and symbol tables — Comparison & intuition

For **Runtime environments, activation records and symbol tables**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Comparison table. Ask what happens when the input is small versus large, structured versus arbitrary, and best-case versus worst-case. Then compare the primary method for Runtime environments, activation records and symbol tables with its nearest alternative. The winning method is not always the asymptotically fastest one; constraints, preprocessing, memory and correctness conditions matter.

Worked reasoning. Consider a small exam-sized instance related to Runtime environments, activation records and symbol tables. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

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7. Runtime environments, activation records and symbol tables — Derivation / proof idea

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Derivation mindset. Start from the definition, state the invariant, perform one legal transformation at a time, and finish by checking the result against a limiting case. For Runtime environments, activation records and symbol tables, this is more useful than memorizing a proof verbatim because GATE can alter the input representation while preserving the underlying theorem or invariant.

Worked reasoning. Consider a small exam-sized instance related to Runtime environments, activation records and symbol tables. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

FIRST/FOLLOW drive predictive parsing

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7. Runtime environments, activation records and symbol tables — Problem-solving workflow

For **Runtime environments, activation records and symbol tables**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Exam workflow. Read the stem once for the model and once for the requested quantity. Extract symbols and constraints, classify the question, eliminate impossible choices, calculate only what is necessary, then perform a sanity check. For Runtime environments, activation records and symbol tables, a 20–40 second classification step often saves several minutes of algebra.

Worked reasoning. Consider a small exam-sized instance related to Runtime environments, activation records and symbol tables. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

LR shift/reduce decisions use item states

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7. Runtime environments, activation records and symbol tables — Edge cases

For **Runtime environments, activation records and symbol tables**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Edge-case checklist. Test zero/one-sized input, empty structures, repeated values, maximum/minimum representable values, ties, disconnected components, overflow/underflow, and invalid assumptions whenever they make sense for Runtime environments, activation records and symbol tables. Edge cases are where apparently correct reasoning often breaks.

Worked reasoning. Consider a small exam-sized instance related to Runtime environments, activation records and symbol tables. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Three-address code uses temporaries

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7. Runtime environments, activation records and symbol tables — Formula & memory map

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Formula sheet. Relevant formulas and relationships include: FIRST/FOLLOW drive predictive parsing; LR shift/reduce decisions use item states; Three-address code uses temporaries; Live variable: variable is live if used on some future path; Available expression must be computed on every predecessor path. Do not treat these as isolated equations. Write the meaning and units of every variable, and note when the equation stops being valid. A formula earns marks only when the model behind it is correct.

Worked reasoning. Consider a small exam-sized instance related to Runtime environments, activation records and symbol tables. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

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7. Runtime environments, activation records and symbol tables — Unit synthesis

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Examiner perspective. A strong question on Runtime environments, activation records and symbol tables can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Runtime environments, activation records and symbol tables. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

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8. Basic blocks and control-flow graphs — Concept foundation

For **Basic blocks and control-flow graphs**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Worked reasoning. Consider a small exam-sized instance related to Basic blocks and control-flow graphs. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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8. Basic blocks and control-flow graphs — Deep dive

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8. Basic blocks and control-flow graphs — Worked example

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Comparison table. Ask what happens when the input is small versus large, structured versus arbitrary, and best-case versus worst-case. Then compare the primary method for Basic blocks and control-flow graphs with its nearest alternative. The winning method is not always the asymptotically fastest one; constraints, preprocessing, memory and correctness conditions matter.

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Topic summary. By the end of this section you should be able to define the core terms of Basic blocks and control-flow graphs, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.

8. Basic blocks and control-flow graphs — Derivation / proof idea

For **Basic blocks and control-flow graphs**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

The central idea is to build a chain from representation to operation to consequence. In **Basic blocks and control-flow graphs**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Derivation mindset. Start from the definition, state the invariant, perform one legal transformation at a time, and finish by checking the result against a limiting case. For Basic blocks and control-flow graphs, this is more useful than memorizing a proof verbatim because GATE can alter the input representation while preserving the underlying theorem or invariant.

Worked reasoning. Consider a small exam-sized instance related to Basic blocks and control-flow graphs. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

FIRST/FOLLOW drive predictive parsing

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8. Basic blocks and control-flow graphs — Problem-solving workflow

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Exam workflow. Read the stem once for the model and once for the requested quantity. Extract symbols and constraints, classify the question, eliminate impossible choices, calculate only what is necessary, then perform a sanity check. For Basic blocks and control-flow graphs, a 20–40 second classification step often saves several minutes of algebra.

Worked reasoning. Consider a small exam-sized instance related to Basic blocks and control-flow graphs. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

LR shift/reduce decisions use item states

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8. Basic blocks and control-flow graphs — Edge cases

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Edge-case checklist. Test zero/one-sized input, empty structures, repeated values, maximum/minimum representable values, ties, disconnected components, overflow/underflow, and invalid assumptions whenever they make sense for Basic blocks and control-flow graphs. Edge cases are where apparently correct reasoning often breaks.

Worked reasoning. Consider a small exam-sized instance related to Basic blocks and control-flow graphs. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Three-address code uses temporaries

GATE traps. Common mistakes around Basic blocks and control-flow graphs include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

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8. Basic blocks and control-flow graphs — Formula & memory map

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Formula sheet. Relevant formulas and relationships include: FIRST/FOLLOW drive predictive parsing; LR shift/reduce decisions use item states; Three-address code uses temporaries; Live variable: variable is live if used on some future path; Available expression must be computed on every predecessor path. Do not treat these as isolated equations. Write the meaning and units of every variable, and note when the equation stops being valid. A formula earns marks only when the model behind it is correct.

Worked reasoning. Consider a small exam-sized instance related to Basic blocks and control-flow graphs. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

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8. Basic blocks and control-flow graphs — Unit synthesis

For **Basic blocks and control-flow graphs**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Examiner perspective. A strong question on Basic blocks and control-flow graphs can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Basic blocks and control-flow graphs. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Available expression must be computed on every predecessor path

GATE traps. Common mistakes around Basic blocks and control-flow graphs include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

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9. Data-flow analyses: constant propagation and liveness — Concept foundation

For **Data-flow analyses: constant propagation and liveness**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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High-yield formula / relation

FIRST/FOLLOW drive predictive parsing

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9. Data-flow analyses: constant propagation and liveness — Deep dive

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Examiner perspective. A strong question on Data-flow analyses: constant propagation and liveness can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Data-flow analyses: constant propagation and liveness. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

LR shift/reduce decisions use item states

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9. Data-flow analyses: constant propagation and liveness — Worked example

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Worked reasoning. Consider a small exam-sized instance related to Data-flow analyses: constant propagation and liveness. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Three-address code uses temporaries

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9. Data-flow analyses: constant propagation and liveness — GATE traps

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High-yield formula / relation

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9. Data-flow analyses: constant propagation and liveness — Comparison & intuition

For **Data-flow analyses: constant propagation and liveness**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Comparison table. Ask what happens when the input is small versus large, structured versus arbitrary, and best-case versus worst-case. Then compare the primary method for Data-flow analyses: constant propagation and liveness with its nearest alternative. The winning method is not always the asymptotically fastest one; constraints, preprocessing, memory and correctness conditions matter.

Worked reasoning. Consider a small exam-sized instance related to Data-flow analyses: constant propagation and liveness. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

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9. Data-flow analyses: constant propagation and liveness — Derivation / proof idea

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Derivation mindset. Start from the definition, state the invariant, perform one legal transformation at a time, and finish by checking the result against a limiting case. For Data-flow analyses: constant propagation and liveness, this is more useful than memorizing a proof verbatim because GATE can alter the input representation while preserving the underlying theorem or invariant.

Worked reasoning. Consider a small exam-sized instance related to Data-flow analyses: constant propagation and liveness. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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9. Data-flow analyses: constant propagation and liveness — Edge cases

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9. Data-flow analyses: constant propagation and liveness — Formula & memory map

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Formula sheet. Relevant formulas and relationships include: FIRST/FOLLOW drive predictive parsing; LR shift/reduce decisions use item states; Three-address code uses temporaries; Live variable: variable is live if used on some future path; Available expression must be computed on every predecessor path. Do not treat these as isolated equations. Write the meaning and units of every variable, and note when the equation stops being valid. A formula earns marks only when the model behind it is correct.

Worked reasoning. Consider a small exam-sized instance related to Data-flow analyses: constant propagation and liveness. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

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The central idea is to build a chain from representation to operation to consequence. In **Data-flow analyses: constant propagation and liveness**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Examiner perspective. A strong question on Data-flow analyses: constant propagation and liveness can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Data-flow analyses: constant propagation and liveness. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Available expression must be computed on every predecessor path

GATE traps. Common mistakes around Data-flow analyses: constant propagation and liveness include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

Topic summary. By the end of this section you should be able to define the core terms of Data-flow analyses: constant propagation and liveness, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.

10. Local optimization, common subexpression elimination and code generation — Concept foundation

For **Local optimization, common subexpression elimination and code generation**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

The central idea is to build a chain from representation to operation to consequence. In **Local optimization, common subexpression elimination and code generation**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Examiner perspective. A strong question on Local optimization, common subexpression elimination and code generation can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Local optimization, common subexpression elimination and code generation. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

FIRST/FOLLOW drive predictive parsing

GATE traps. Common mistakes around Local optimization, common subexpression elimination and code generation include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

Topic summary. By the end of this section you should be able to define the core terms of Local optimization, common subexpression elimination and code generation, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.

10. Local optimization, common subexpression elimination and code generation — Deep dive

For **Local optimization, common subexpression elimination and code generation**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

The central idea is to build a chain from representation to operation to consequence. In **Local optimization, common subexpression elimination and code generation**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Examiner perspective. A strong question on Local optimization, common subexpression elimination and code generation can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Local optimization, common subexpression elimination and code generation. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

LR shift/reduce decisions use item states

GATE traps. Common mistakes around Local optimization, common subexpression elimination and code generation include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

Topic summary. By the end of this section you should be able to define the core terms of Local optimization, common subexpression elimination and code generation, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.

10. Local optimization, common subexpression elimination and code generation — Worked example

For **Local optimization, common subexpression elimination and code generation**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

The central idea is to build a chain from representation to operation to consequence. In **Local optimization, common subexpression elimination and code generation**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Examiner perspective. A strong question on Local optimization, common subexpression elimination and code generation can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Local optimization, common subexpression elimination and code generation. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Three-address code uses temporaries

GATE traps. Common mistakes around Local optimization, common subexpression elimination and code generation include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

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10. Local optimization, common subexpression elimination and code generation — GATE traps

For **Local optimization, common subexpression elimination and code generation**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Examiner perspective. A strong question on Local optimization, common subexpression elimination and code generation can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Local optimization, common subexpression elimination and code generation. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Live variable: variable is live if used on some future path

GATE traps. Common mistakes around Local optimization, common subexpression elimination and code generation include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

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10. Local optimization, common subexpression elimination and code generation — Comparison & intuition

For **Local optimization, common subexpression elimination and code generation**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

The central idea is to build a chain from representation to operation to consequence. In **Local optimization, common subexpression elimination and code generation**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Comparison table. Ask what happens when the input is small versus large, structured versus arbitrary, and best-case versus worst-case. Then compare the primary method for Local optimization, common subexpression elimination and code generation with its nearest alternative. The winning method is not always the asymptotically fastest one; constraints, preprocessing, memory and correctness conditions matter.

Worked reasoning. Consider a small exam-sized instance related to Local optimization, common subexpression elimination and code generation. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Available expression must be computed on every predecessor path

GATE traps. Common mistakes around Local optimization, common subexpression elimination and code generation include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

Topic summary. By the end of this section you should be able to define the core terms of Local optimization, common subexpression elimination and code generation, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.

10. Local optimization, common subexpression elimination and code generation — Derivation / proof idea

For **Local optimization, common subexpression elimination and code generation**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

The central idea is to build a chain from representation to operation to consequence. In **Local optimization, common subexpression elimination and code generation**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Derivation mindset. Start from the definition, state the invariant, perform one legal transformation at a time, and finish by checking the result against a limiting case. For Local optimization, common subexpression elimination and code generation, this is more useful than memorizing a proof verbatim because GATE can alter the input representation while preserving the underlying theorem or invariant.

Worked reasoning. Consider a small exam-sized instance related to Local optimization, common subexpression elimination and code generation. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

FIRST/FOLLOW drive predictive parsing

GATE traps. Common mistakes around Local optimization, common subexpression elimination and code generation include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

Topic summary. By the end of this section you should be able to define the core terms of Local optimization, common subexpression elimination and code generation, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.

10. Local optimization, common subexpression elimination and code generation — Problem-solving workflow

For **Local optimization, common subexpression elimination and code generation**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

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Exam workflow. Read the stem once for the model and once for the requested quantity. Extract symbols and constraints, classify the question, eliminate impossible choices, calculate only what is necessary, then perform a sanity check. For Local optimization, common subexpression elimination and code generation, a 20–40 second classification step often saves several minutes of algebra.

Worked reasoning. Consider a small exam-sized instance related to Local optimization, common subexpression elimination and code generation. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

LR shift/reduce decisions use item states

GATE traps. Common mistakes around Local optimization, common subexpression elimination and code generation include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

Topic summary. By the end of this section you should be able to define the core terms of Local optimization, common subexpression elimination and code generation, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.

10. Local optimization, common subexpression elimination and code generation — Edge cases

For **Local optimization, common subexpression elimination and code generation**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

The central idea is to build a chain from representation to operation to consequence. In **Local optimization, common subexpression elimination and code generation**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Edge-case checklist. Test zero/one-sized input, empty structures, repeated values, maximum/minimum representable values, ties, disconnected components, overflow/underflow, and invalid assumptions whenever they make sense for Local optimization, common subexpression elimination and code generation. Edge cases are where apparently correct reasoning often breaks.

Worked reasoning. Consider a small exam-sized instance related to Local optimization, common subexpression elimination and code generation. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Three-address code uses temporaries

GATE traps. Common mistakes around Local optimization, common subexpression elimination and code generation include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

Topic summary. By the end of this section you should be able to define the core terms of Local optimization, common subexpression elimination and code generation, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.

10. Local optimization, common subexpression elimination and code generation — Formula & memory map

For **Local optimization, common subexpression elimination and code generation**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

The central idea is to build a chain from representation to operation to consequence. In **Local optimization, common subexpression elimination and code generation**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Formula sheet. Relevant formulas and relationships include: FIRST/FOLLOW drive predictive parsing; LR shift/reduce decisions use item states; Three-address code uses temporaries; Live variable: variable is live if used on some future path; Available expression must be computed on every predecessor path. Do not treat these as isolated equations. Write the meaning and units of every variable, and note when the equation stops being valid. A formula earns marks only when the model behind it is correct.

Worked reasoning. Consider a small exam-sized instance related to Local optimization, common subexpression elimination and code generation. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Live variable: variable is live if used on some future path

GATE traps. Common mistakes around Local optimization, common subexpression elimination and code generation include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

Topic summary. By the end of this section you should be able to define the core terms of Local optimization, common subexpression elimination and code generation, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.

10. Local optimization, common subexpression elimination and code generation — Unit synthesis

For **Local optimization, common subexpression elimination and code generation**, start with precise definitions rather than memorized keywords. In GATE, a definition is often embedded inside a numerical or conceptual statement, so identify the object, its assumptions, its allowed operations, and the property that must remain invariant. A strong solution separates what is always true from what is true only under a stated condition.

The central idea is to build a chain from representation to operation to consequence. In **Local optimization, common subexpression elimination and code generation**, first choose the canonical representation, then track how an operation changes the representation, and finally check the boundary cases. This is why two formulas that look similar can give different answers: one may assume independence, a balanced tree, a valid pipeline schedule, a particular addressing mode, or a specific language class. Always write the assumption before applying a shortcut.

Examiner perspective. A strong question on Local optimization, common subexpression elimination and code generation can combine two chapters. For example, a representation may determine an algorithmic complexity, or a protocol rule may determine a probability or queueing calculation. Practice identifying the dependency rather than solving each chapter in isolation.

Worked reasoning. Consider a small exam-sized instance related to Local optimization, common subexpression elimination and code generation. Suppose the input has three interacting components. Write the state after each operation instead of jumping to the final answer. For each transition ask: what changed, what did not change, and what invariant can be checked? If two candidate answers remain, use a counterexample. This method is reliable for both numerical questions and multiple-select conceptual questions.

High-yield formula / relation

Available expression must be computed on every predecessor path

GATE traps. Common mistakes around Local optimization, common subexpression elimination and code generation include confusing necessary and sufficient conditions, ignoring a boundary value, using an average where a weighted average is required, applying an asymptotic result to a small constant case, or assuming that an implementation detail is part of an abstract specification. Another frequent trap is silently changing the model midway through a calculation. Mark the exact model before solving.

Topic summary. By the end of this section you should be able to define the core terms of Local optimization, common subexpression elimination and code generation, derive or justify the main relationship, solve a small instance without notes, explain one counterexample to a tempting false statement, and recognize the question pattern in which the topic is usually tested. Revision should compress the topic into one diagram, three rules, two formulas and one representative problem.